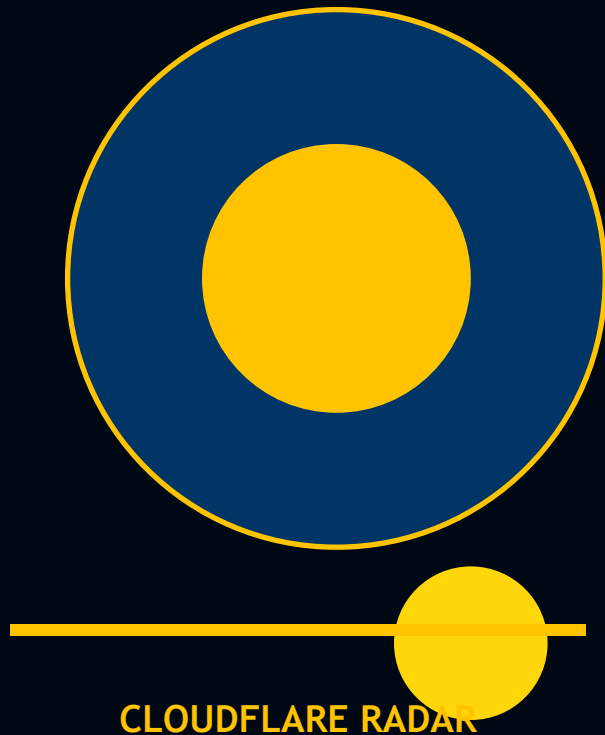


Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats

2024 Annual Report

Data from Cloudflare's global network | radar.cloudflare.com



radar.cloudflare.com

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Year-over-Year Traffic Growth

3.3x

Starlink Traffic Surge in 2024

Key Insights

- Global traffic grew 17% in 2024, driven by commerce, entertainment, and communication demands
- SpaceX Starlink achieved a remarkable 3.3x traffic surge as satellite internet reached underserved regions
- Traffic trend normalized from mid-January to exclude post-holiday dips
- Starlink scaled relative to peak observed request volume; not yet available globally
- Satellite connectivity is closing the digital divide in remote regions worldwide

Google Leads Top 10 Services as Generative AI Rises

RANK SERVICE

- | | |
|-----|-----------|
| #1 | Google |
| #2 | Facebook |
| #3 | Apple |
| #4 | TikTok |
| #5 | AWS |
| #6 | Microsoft |
| #7 | Instagram |
| #8 | YouTube |
| #9 | Amazon |
| #10 | WhatsApp |

Key Insight



Continued its rapid rise throughout 2024, becoming one of the fastest-growing categories

Methodology

Rankings based on aggregate DNS resolver data from Cloudflare's 1.1.1.1 service

Notable Trends

Google leads for 2nd consecutive year · Metaverse category receded in 2024

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-Quantum Encryption (TLS 1.3 Traffic)

Chrome 124 enabled post-quantum key agreement by default on April 17, 2024, triggering a rapid jump from ~2% to ~13%. Firefox and Safari are following with rollouts.

21%

HTTP/3 Adoption

41%

Mobile Device Traffic

Top API Languages

Go ~12%

Node.js #2

Python 9.6%

Java 7.4% | .NET 3.6%

IPv6 adoption reached 28% of dual-stack traffic in 2024. Google leads search engine referrals at 89%, followed by Yandex (3.1%), Baidu (2.7%), and Bing (2.6%).

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Disruptions
in 2024

Government Directed

Cable Cut

Unknown

Power Outage

Technical Problem

Military Action

41

16

16

9

9

Key Findings

Government-Directed Shutdowns

116+ incidents — the leading cause, often linked to elections, protests, or political events

Cable Cuts

41 incidents from undersea and terrestrial cable damage

Power Outages

16 incidents — infrastructure dependency on power grids

Tracked via Cloudflare Radar Outage Center using real-time traffic data

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

Most Attacked Sector

Gambling & Games

Highest attack volume by sector

68%

of global bot traffic from
top 10 countries

Log4Shell Still Active

CVE-2021-44228 (Log4j) disclosed December 2021 remains actively exploited 3+ years later — a persistent reminder that critical vulnerabilities endure

Email Security:

4.3% of processed emails were malicious · 43% contained deceptive links · Top malicious TLD: .bar
· RPKI valid routes +6.4%

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global traffic +17% YoY; Starlink 3.3x surge marks satellite connectivity as a transformative force for underserved regions

2

Post-Quantum Momentum

Encryption share jumped from 2% to 13% of TLS 1.3 in months — Chrome 124 proved one update can reshape security at Internet scale

3

Connectivity Fragility

225 disruptions in 2024; government-directed shutdowns (116+) now rival infrastructure failures as leading threats to Internet continuity

4

Persistent Security Threats

6.5% of traffic mitigated; Log4j still exploited years after disclosure; Gambling & Games most targeted; 4.3% of emails malicious

5

Evolving Developer Ecosystem

Go leads API clients at ~12%; generative AI reshapes service demand; AI-native architectures accelerating across the developer toolchain